

Lesson 6: File Handling and Exceptions

Activity 1: Exception Handling Practice (50 points)

Filename: Surname_06_FileHandlingAndExceptions_Practice

Objective

Create a program that demonstrates various exception handling techniques including try-catch-finally, custom exceptions, and exception propagation.

Requirements

- Create a custom checked exception: **InvalidGradeException**
 - Thrown when grade is not between 0 and 100
 - Include the invalid grade value in the exception message
- Create a **GradeCalculator** class with:
 - Method: `double calculateAverage(double[] grades)` throws `InvalidGradeException`
 - Validate each grade (0-100)
 - Throw `InvalidGradeException` if any grade is invalid
- Create a **main** method that:
 - Accepts 5 grades from user input
 - Uses try-catch to handle `InvalidGradeException`
 - Uses try-catch to handle `InputMismatchException` (invalid input)
 - Uses finally block to display "Process complete"
 - Displays the average if all grades are valid

Expected Output Example

```
Enter grade 1: 85
Enter grade 2: 92
Enter grade 3: 105
Error: Invalid grade 105.0. Grade must be between 0 and 100.
Process complete.
```

Activity 2: Contact Book File System

Filename: Surname_07_FileHandlingAndExceptions_ContactBookFileSystem

Objective

Build a contact management system that stores and retrieves contact information from a text file.

Requirements

- Create a **Contact** class with:
 - Attributes: name, phone, email
 - Constructor and getters
 - toString() method for display
- Create a **ContactBook** class with methods:
 - void addContact(Contact contact)
 - * Append contact to contacts.txt file
 - * Format: name,phone,email
 - void displayAllContacts()
 - * Read and display all contacts from file
 - Contact searchContact(String name)
 - * Search by name, return Contact or null
- Use try-with-resources for all file operations
- Handle FileNotFoundException and IOException separately
- Create a menu-driven interface (Add, Display, Search, Exit)

Expected Output Example

```
=== Contact Book ===
1. Add Contact
2. Display All Contacts
3. Search Contact
4. Exit
Choice: 1
Enter name: John Doe
Enter phone: 555-1234
Enter email: john@email.com
Contact added successfully!
```

